

Agenda

- ① Iterative control statements
- ② while loop
- ③ Practice problems
- ④ do-while loop
- ⑤ for loop
- ⑥ break
- ⑦ continue

Iterative Control Statements (loop)

- ① while
- ② do while
- ③ for

```
int main()  
{
```

```
    _____  
    _____  
    _____  
    _____  
    while(—)  
    {  
        _____  
        _____  
    }  
    _____  
    _____  
    _____
```

```
}
```

while

while (condition)

{

==
==
==

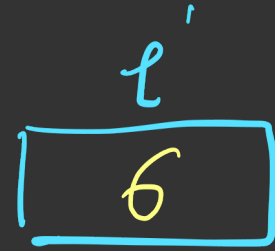
}

while
block

F

Write a program to print MySirG
5 times on the screen.

```
int main()  
{  
    int i=1;
```



```
    while (i <= 5)  
    {  
        printf("MySirG");  
        i++;  
    }
```

```
}
```

Write a program to print first 10 natural numbers.

1 2 3 4 5 6 7 8 9 10

```
int main()
```

```
{
```

```
    int i=1;
```

```
    while(i<=10)
```

```
    {
```

```
        printf("%d ", i);
```

```
        i++;
```

```
    }
```

```
}
```



Write a program to print first
N natural numbers.

Find Output of the program

```
int main()  
{
```

```
    int i = 0;
```

```
    while (i <= 6)  
    {
```

```
        if (i % 2)
```

```
            i = i - 3;
```

```
        else
```

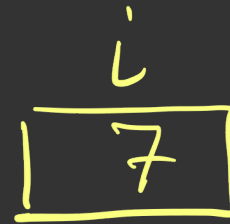
```
            i = i + 5;
```

```
        printf("%d ", i);
```

```
    }
```

```
    printf("%d", i);
```

```
}
```



$0 \% 2 = 0$

5 2 7 7

$-5 \% 2 = -1$

$5 \% -2 = 1$

$-5 \% -2 = -1$

$5 \% 2 = 1$

Entry Control loop

```
while (condition)
```

```
{  
    ==  
    ==  
}
```

```
int i = 1;
```

```
while (i <= 10)
```

```
{  
    printf("%d", i);  
    i++;  
}
```

```
1 <= 10 T  
2 <= 10 T  
3 <= 10 T  
...
```

```
10 <= 10 T  
11 <= 10 F
```

Exit Control loop

```
do
```

```
{  
    ==  
    ==  
}
```

```
while (condition);
```

```
int i = 1;
```

```
do
```

```
{  
    printf("%d", i);  
    i++;  
}
```

```
while (i <= 10);
```

```
2 <= 10 T  
3 <= 10 T  
4 <= 10 T  
...
```

```
10 <= 10 T  
11 <= 10 F
```

Entry Control loop

```
for ( ; ; )
```

```
{  
    ==  
    ==  
}
```

```
int i;
```

```
for (i = 1; i <= 10; i++)
```

```
{  
    printf("%d", i);  
}
```

```
1 <= 10 T  
2 <= 10 T  
3 <= 10 T  
...
```

```
10 <= 10 T  
11 <= 10 F
```

```
int main()           while ( ) error
{
    int i=1;

    for ( ; True ; )
    {
        printf("%d ", i);
        i++;
    }
}
```

Find Output of the program

```
int main()  
{
```

```
    int a=2, b=-1, c;
```

```
    do  
    {
```

```
        c = a + b + 1;
```

```
        printf("%d", c);
```

```
        a = b + 1;
```

```
        b = a + 1;
```

```
    } while (c < 10);
```

```
}
```



2 2 6 10

Find output of the program

```
int main()
```

```
{
```

```
    int i, j, k;
```

```
    for (i=0; i<3; i++)
```

```
    {
```

```
        j = i*2 + 1;
```

```
        k = j*3 - 2;
```

```
    }
```

```
    printf("%d %d %d", i, j, k);
```

```
}
```

i	j	k
3	5	13

3 5 13

break

```
while(condition)
{
```

```
    ==
    break;
    ==
}
```

```
int i = 1;
while(i <= 100)
{
```

```
    if( )
        break;
    == i++;
}
```

- **break** is a keyword
- It can be used in the body of loop or in the body of switch.
- When **break** encounters loop terminates and control move out of the loop body.



Write a program which asks user to enter an even number, computer will give user at most 3 chances, if user failed to answer in any of the chances, "Game Over" message should be displayed on the screen, otherwise "You Win" message should be displayed and game stops immediately.

Write a program to add numbers entered by user. User can enter any number of numbers until he enters 0.

S
95

x
40

$$\underline{S = S + x}$$

Continue

```
while (condition)
{
    //
    continue;
    //
}
```

- continue is a keyword
- continue can only be used in the body of loop.
- continue transfers the control to the next iteration.